Automated Textile Thread/Needle Insertion System

Complete Technical Manual: System Prerequisites, Installation & Operator Functionalities

1. System Requirements & Software to Install

Before starting the application on Windows, verify that the following prerequisites are installed:

Software / Tool	Required Version	Verification Command	Purpose
Python	3.11 or 3.12 (64-bit)	python --version	FastAPI, OpenCV CV Pipeline, Kinematics Simulator
Node.js	LTS (v20+ or v24+)	node --version	JavaScript runtime for Angular toolchain
npm	v10+ or v11+	npm --version	Frontend package manager
Angular CLI	v19+ or v22+	ng version	Angular development server and compiler
Git	v2.40+	git --version	Source control management

2. Installation & Running Commands

Backend Installation (PowerShell / Command Prompt):

```
cd C:\Users\sathu\Desktop\App\backend
python -m venv venv
.\venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
pip install fastapi uvicorn opencv-python numpy scipy pyserial websockets sqlalchemy alembic python-multipart
pydantic httpx pytest
uvicorn app.main:app --reload --host 127.0.0.1 --port 8000
```

Frontend Installation (PowerShell / Command Prompt):

```
cd C:\Users\sathu\Desktop\App\frontend
npm install
npm install @angular/material @angular/cdk zone.js
npm start
```

3. Detailed Functional Guide

• Upload Board Image

Allows the operator to upload any high-resolution physical textile board photo (.png, .jpg, .bmp). The image is safely stored in backend/storage/boards/ with unique IDs and pixel dimensions indexed in the database.

• Detect Holes (OpenCV Computer Vision)

Runs an advanced computer vision pipeline (Grayscale -> Gaussian blur -> Adaptive thresholding -> Morphological filtering -> Circularity filtering -> Sub-pixel centroid moments). Detects every actual physical hole without making rigid rectangular grid assumptions.

• Confirm Hole Map

Locks and verifies the detected hole map for production. Prevents accidental modifications and ensures coordinate accuracy before triggering physical or simulated CNC needle movements.

• Excel-Style Hole Labeling (A1..ZZ20)

Automatically numbers detected holes into rows and columns using scalable Excel naming (A1..A20, B1..B20, Z1..Z20, AA1..AA50, AB1, etc.). Supports 5,000 to 10,000+ holes with zero label collisions.

• Manual Hole Selection & Deselection

The operator can click directly on any hole on the canvas board image to select or deselect it. Selected holes turn Electric Blue and are added to the execution sequence queue.

• Select All & Clear Selection

Allows the operator to queue all detected holes with a single click ('Select All Holes') or clear the entire queue ('Clear Selection') to start fresh.

• Start Job

Initiates sequential CNC execution. The virtual machine head moves smoothly across X, Y at Safe Z height (25mm), lowers the needle downstroke (5mm), actuates thread insertion, and captures a live camera frame for optical verification.

• Pause Job & Resume Job

Provides real-time machine execution control. Pausing freezes all stepper motion and holds position. Resuming continues seamlessly from the exact current hole without skipping or repeating.

• Stop Job

Safely aborts the active insertion job, returns the toolhead to Safe Z height, and marks the job as STOPPED.

• Live Camera Inspection & Fiber Verification

Streams live video from a connected USB camera (or high-definition synthetic inspection feed). Upon needle actuation, OpenCV analyzes fiber density, texture variance, and edge deviation to verify thread presence, returning THREAD_DETECTED or THREAD_NOT_DETECTED.

• Hole Retry Workflow (Green / Red Status)

Holes that pass verification turn Emerald Green (SUCCESS). If a hole fails, it turns Crimson Red (FAILED). The operator can select the failed hole and click RETRY to re-position and re-insert immediately.

• HOME Button (G28 Reference Homing)

Drives the CNC axes to the physical limit switches at (0, 0, 25mm). Establishes the real-world hardware coordinate origin so all subsequent millimeter coordinates are mathematically accurate.

• CALIBRATE Button (Planar Homography)

Calculates a 3x3 perspective transformation matrix mapping camera pixels (px) to physical CNC millimeters (mm) using 4 corner reference points, validating sub-millimeter reprojection accuracy.

• Hole Search & 3D Coordinates Inspector

Allows searching by label (e.g. A1, B3) to instantly pan, zoom, and display all 3 coordinate systems: Image Pixels (X, Y px), Board Layout (X, Y mm), and Machine CNC (X, Y, Z mm).

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4. Operator Control Quick Reference

Control / Action	Description	Visual Indicator
Click Hole	Select/deselect hole for insertion	Turns **Electric Blue**
Detect Holes	OpenCV hole detection & Excel labeling	Renders Amber dots + A1..N20
START JOB	Starts 3D movement & thread insertion	Current hole pulses **Cyan**
Success Result	Thread confirmed by camera check	Turns **Emerald Green**
Failure Result	Thread missing or low confidence	Turns **Crimson Red**
RETRY	Re-runs insertion on failed hole	Re-executes needle stroke
HOME	Re-zeros machine to (0,0,25mm)	Position resets to 0.00, 0.00
E-STOP	Emergency software halt	Machine status = STOPPED